



Amazon ElastiCache

Lecture Slides

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Good luck on your exam!

What is In-Memory Data Store?

What is Caching?

Caching is the process of storing data in a cache.

A cache is a **temporary storage** area.

Caches are optimized for fast retrieval with the trade-off that data is not durable.

What is In-Memory Data Store?

When data is stored In-Memory (think of RAM).

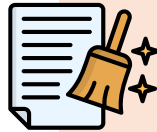
The trade-off is high volatility (low durability, risk of data loss), but **access** to data is **very fast**.



Introduction to ElastiCache

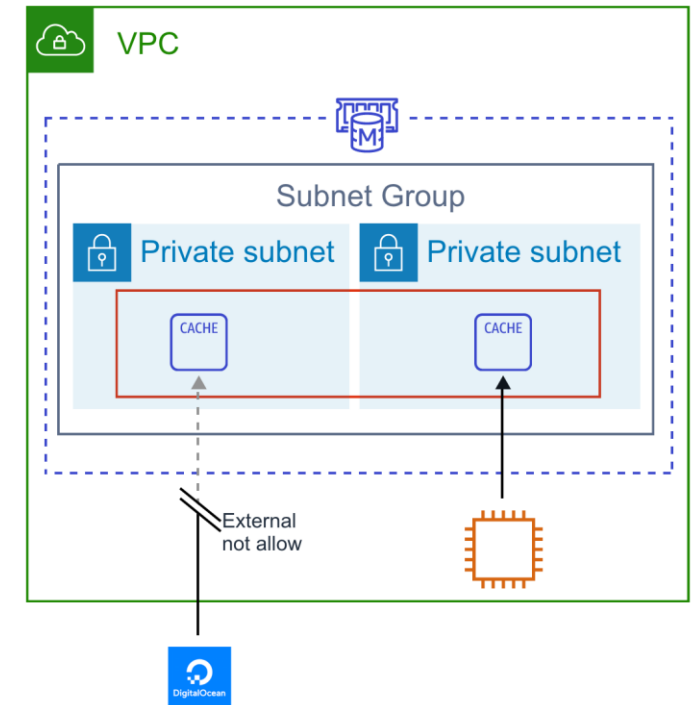


ElastiCache is a **fully managed in-memory datastore** for either open-source data stores Memcached or Redis.



ElastiCache is intended to cache data or HTML fragments to greatly improve response times in 10s to 100s of milliseconds

- ElastiCache can **only accessible by resources in the same VPC**.
 - This is ensure low-latency
- ElastiCache can be deployed in multiple AZs for high availability
- ElastiCache can be deployed on-premise via AWS Outposts
- ElastiCache can use RBAC for Redis 6.0+ so you manage user access via the AWS Management Console
- ElastiCache can be replicated cross-region via ElastiCache Global Datastores
- You can reserve nodes to save money when using ElastiCache Standard
- ElastiCache can be automated to perform backups of data stores.



ElastiCache – Deployment Options

ElastiCache has two deployment options:

- ElastiCache (standard mode directly managing nodes in a cluster)
- ElastiCache Serverless

	Serverless	Standard
Use Case	Unpredictable workloads	Predictable Workloads
Management	Automatically scales	Customer manages cluster and nodes
Billing	• Data Stored • ElastiCache Processing Units (ECPUs)	• Based on number of nodes • Based on node type (size of node)

*Standard can be deployed on-premise
via AWS Outposts*

ElastiCache – Caching Comparison

Memcached is generally preferred for caching HTML fragments. Memcached is a simple key/value store. The trade-off to being simple is that it's very fast

Redis can perform many different kinds of operations on your data. It's very good for leaderboards, and keep track of unread notification data. It's very fast, but **arguably** not as fast as Memcached.



Don't google **"Memcache vs Redis"** unless you want to read endless arguments as if people are arguing "Kirk vs Picard"



		
Sub-millisecond latency	Yes	Yes
Developer ease of use	Yes	Yes
Data partitioning	Yes	Yes
Support for a broad set of programming languages	Yes	Yes
Advanced data structures	—	Yes
Multithreaded architecture	Yes	—
Snapshots	—	Yes
Replication	—	Yes
Transactions	—	Yes
Pub/Sub	—	Yes
Lua scripting	—	Yes
Geospatial support	—	Yes